

A disproof of Conjecture 00000002747

The capacity of $M_m(M_n)$ is not $(m-1)n^2 + n$

1 The conjecture and its reading

Conjecture 00000002747 reads: *Definition: Capacity: the maximal dimension of a matrix space on which the determinant vanishes. Conjecture: The higher-order Flanders–Atkinson capacity theorem: the capacity of $M_m(M_n)$ is $(m-1)n^2 + n$, and the classification of capacity spaces is a single orbit of linear spaces.*

Reading. Fix any field K and integers $m, n \geq 1$.

- $M_m(M_n) = M_m(M_n(K))$ is the algebra of $m \times m$ matrices whose entries are $n \times n$ matrices over K .
- Its determinant is the determinant of the $mn \times mn$ scalar matrix obtained by forgetting the block structure, i.e. via the canonical isomorphism $M_m(M_n(K)) \cong M_{mn}(K)$, $A \mapsto (A_{ij}(k, l))_{(i,k),(j,l)}$.
- A *singular space* is a K -linear subspace $V \subseteq M_m(M_n(K))$ with $\det A = 0$ for all $A \in V$.
- The *capacity* $\text{cap}(m, n)$ is the supremum (a maximum, since dimensions are bounded by $m^2 n^2$) of $\dim_K V$ over singular spaces V .

The conjecture is a conjunction. We refute its first clause, “ $\text{cap}(m, n) = (m-1)n^2 + n$ for all $m, n \geq 1$ ”. This refutes the conjunction whatever the precise meaning of the second (classification) clause; that clause is not analysed.

2 The counterexamples

Lemma 1 ($m = n = 1$). $\text{cap}(1, 1) = 0$, while $(1-1) \cdot 1^2 + 1 = 1$.

Proof. $M_1(M_1(K))$ is one-dimensional and the determinant of a 1×1 matrix is its entry. So a singular space contains only the zero matrix, and its dimension is 0. $\square$

Lemma 2 (zero-row space). Let $m, n \geq 1$, $N = mn$, and fix a scalar row index (i_0, k_0) . The space V_0 of block matrices whose underlying $N \times N$ matrix has zero row (i_0, k_0) is singular and $\dim_K V_0 = N^2 - N$. Hence $\text{cap}(m, n) \geq N^2 - N$.

Proof. A square matrix with a zero row has determinant 0, so V_0 is singular. Under the block isomorphism, V_0 corresponds to the kernel of the row map $B \mapsto B_{(i_0, k_0), \bullet}$, from $M_N(K)$ onto K^N . This map is surjective, so by rank–nullity its kernel has dimension $N^2 - N$. $\square$

Lemma 3 (arithmetic). If $m, n \geq 1$ and $mn \geq 3$, then $(m-1)n^2 + n < (mn)^2 - mn$.

Proof. Write $m = a + 1$ and $n = b + 1$. The claim is equivalent to $(a + 1)^2(b + 1) > ab + 2a + 2$, i.e. to $a^2b + a^2 + ab + b \geq 2$. The hypothesis $mn \geq 3$ means $ab + a + b \geq 2$. If $a = 0$ then $b \geq 2$; if $b = 0$ then $a \geq 2$, so $a^2 \geq 4$; if $a, b \geq 1$ then $a^2b + b \geq 2$. $\square$

Theorem 4. *For every field K :*

1. $\text{cap}(1, 1) = 0 \neq 1$;
2. for all $m, n \geq 1$ with $mn \geq 3$, $\text{cap}(m, n) > (m - 1)n^2 + n$; for example $\text{cap}(2, 2) \geq 12 > 6$.

So the capacity formula fails for $(m, n) = (1, 1)$ and for every $m, n \geq 1$ with $mn \geq 3$. It can hold only at $(m, n) \in \{(1, 2), (2, 1)\}$, which we do not need. In particular, the conjunction stated in the conjecture is false.

Remark 5. The theorem uses only the lower bound $\text{cap}(m, n) \geq N^2 - N$ (and the trivial case $m = n = 1$). No upper bound on singular spaces of size $N \geq 2$ is needed or claimed.

3 Relation to the earlier submission

PR #198 by orionsheep was closed unmerged. Its mathematics (the cases $m = n = 1$ and $m = n = 2$) is the same as here. The reason for closing was that “the main Lean theorem is vacuous arithmetic with none of the conjecture’s objects in Lean”: there was “no matrix space, determinant, or linear subspace in Lean (theorems are $0 \neq 1$ and $12 > 6$); the decisive facts — $\det(\lambda) = \lambda$ on scalars and the 12-dimensional row-zero subspace being identically singular — are prose only.”

The present Lean file defines the following objects:

- the block algebra `BlockMat K m n =  $M_m(M_n(K))$` ;
- its determinant `blockDet`, through Mathlib’s block isomorphism `Matrix.comp`;
- singular subspaces as `Submodules`;
- `capacity` as the supremum of their `finrank`.

It proves both facts that the reviewer flagged as prose-only. It also proves the failure for every $m, n \geq 1$ with $mn \geq 3$, not only at $m = n = 2$.

4 Lean formalization

The file is `lean/Conjecture2747/Basic.lean` (namespace `C2747`). It imports only Mathlib.

- Definitions:
 - `BlockMat K m n := Matrix (Fin m) (Fin m) (Matrix (Fin n) (Fin n) K)`;
 - `blockDet A := (Matrix.comp _ _ _ _ K A).det`;
 - `IsSingularSpace V for V : Submodule K (BlockMat K m n): every member has blockDet 0`;
 - `capacity K m n := sSup {d |  $\exists V, \text{IsSingularSpace } K V \wedge \text{finrank } K V = d$ }`.
- Bounds on the supremum: `finrank_le_capacity` and `capacity_le`.

- `capacity_one_one` : `capacity K 1 1 = 0`.
- Zero-row space:
 - `rowZeroSpace` (the image of `ker (rowMap (i0,k0))` under the inverse of `Matrix.compLinearEquiv`);
 - `rowZeroSpace_singular`;
 - `finrank_rowZeroSpace`: the dimension is $(mn)(mn) - mn$;
 - `capacity_ge`: the capacity is at least $(mn)(mn) - mn$.
- Failure of the formula:
 - `formula_lt_capacity`: for $m, n \geq 1$ and $mn \geq 3$, $(m-1)n^2 + n < \text{capacity K m n}$;
 - `capacity_two_two`: $12 \leq \text{capacity K 2 2}$.
- Main theorem `conjecture_false` (`Classif :  $\mathbb{N} \rightarrow \mathbb{N} \rightarrow \text{Prop}$`):
$$\neg \forall m n, 1 \leq m \rightarrow 1 \leq n \rightarrow (\text{capacity K m n} = (m-1)n^2 + n \wedge \text{Classif m n}).$$

It holds for an arbitrary field K and an arbitrary formalization `Classif` of the classification clause.

- Axiom audit: `#print axioms` for `C2747.conjecture_false`, `C2747.formula_lt_capacity` and `C2747.capacity_two_two` reports `[propext, Classical.choice, Quot.sound]`. There is no `sorry`, no `decide` and no `native_decide`.